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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Social Networks (course)



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Course
outline

How does an
NPTEL
online
course
work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 : Assignment

Your last recorded submission was on 2023-09-12, 00:00 Due date: 2023-09-13, 23:59 IST.

1) A cascade cannot be complete in a network with threshold of adoption = q , if there exist a cluster with density **1 point**

- ☐ lesser than q
- ☒ greater than q
- ☐ lesser than $(1 - q)$
- ☐ greater than $(1 - q)$

2) Let 'x' be a node in a graph. Suppose that a 'p' fraction of the neighbors of 'x' have behavior A, and a $(1 - p)$ fraction have behavior B; that is, if 'x' has 'd' neighbors, then pd adopt A and $(1 - p)d$ adopt B, as shown in the following Figure. Behavior A has a payoff of 'a' and behavior **1 point**



Week 7 ()

- Lecture 87 - We Follow (unit? unit=121&less on=122)
- Lecture 88 - Why do we Follow? (unit? unit=121&less on=123)
- Lecture 89 - Diffusion in Networks (unit? unit=121&less on=124)
- Lecture 90 - Modeling Diffusion (unit? unit=121&less on=125)
- Lecture 91- Modeling Diffusion (Continued) (unit? unit=121&less on=126)
- Lecture 92 - Impact of Communities on Diffusion (unit? unit=121&less on=127)
- Lecture 93 - Cascade and Clusters (unit? unit=121&less on=128)
- Lecture 94 - Knowledge, Thresholds and the Collective Action (unit? unit=121&less on=129)

B has a payoff of 'b'. Then A is a better choice for 'x' if

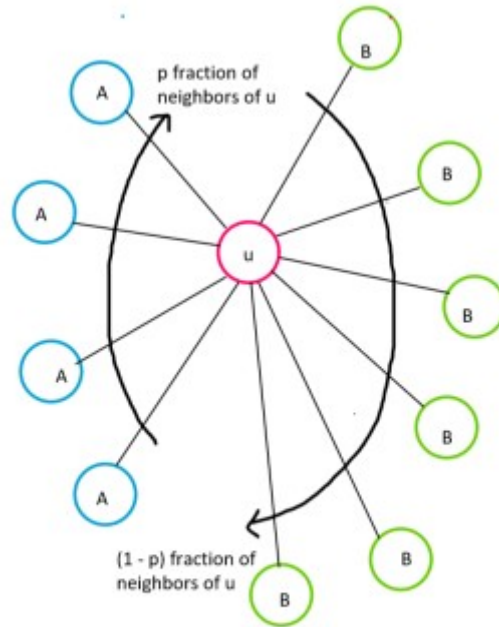


Figure 13: Graph H

- ☐ $p \geq a/b$
- ☐ $p \geq b/a$
- ☐ $p \geq a/(a + b)$
- ☒ $p \geq b/(a + b)$

3) Find the cluster density of the blue color nodes shown in the given figure 14.

1 point

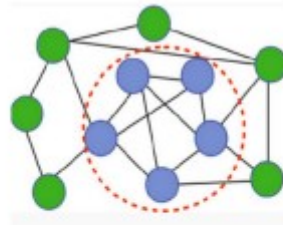


Figure 14: Collective Action

- ☐ 1/2
- ☐ 2/3
- ☐ 3/4
- ☒ 3/5

4) Which of the following are valid factors for influence diffusion?

1 point

- ☒ Payoff
- ☐ Key people
- ☐ Both Payoff and key people
- ☐ Neither Payoff nor key people



● Lecture 95 - An Introduction to the Programming Screenshot (Coding 4 major ideas) (unit? unit=121&lesson=130)

● Lecture 96 - The Base Code (unit? unit=121&lesson=131)

● Lecture 97 - Coding the First Big Idea - Increasing the Payoff (unit? unit=121&lesson=132)

● Lecture 98 - Coding the Second Big Idea - Key People (unit? unit=121&lesson=133)

● Lecture 99 - Coding the Third Big Idea- Impact of Communities on Cascades (unit? unit=121&lesson=134)

● Lecture 100 - Coding the Fourth Big Idea - Cascades and Clusters (unit? unit=121&lesson=135)

● Week 7 Feedback Form: Social Networks (unit?)

5) Consider the basic coordination based game theoretical model for cascading behavior. The payoffs associated with behaviours 'A' and 'B' are a and b respectively. Consider two friends x and y adopting behaviours 'A' and 'B' respectively. Assuming that x and y have no other friends than each other, the total payoff which they receive is **1 point**

- ☐ ab
- ☒ $a + b$
- ☐ a/b
- ☐ $b/(a + b)$

6) Given a network as shown in the following Figure, assume that initially every node in this network has adopted behavior B. Next, a new behavior A is introduced in the network and the nodes 1 and 3 are the initial adopters of this behavior A, i.e., nodes 1 and 3 now have adopted behavior A and the rest of the nodes have adopted behavior B. The payoff associated with A is $a = 3$ and the payoff associated with B is $b = 2$. After the introduction of this new behavior A in the network, all the nodes will start weighing their options and might change their behavior. This leads to a cascade in the network. When the cascade ends, which all are the nodes who have adopted the behavior A. **1 point**

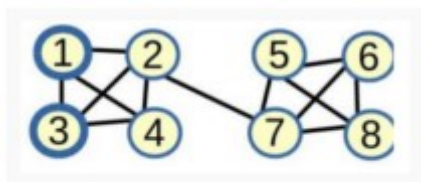


Figure 15: Cascade

- ☐ 1, 3, 2
- ☒ 1, 3, 2, 4
- ☐ 1, 3, 2, 4, 5
- ☐ 1, 3

7) Let us consider a model for collective action shown in Figure 16, where the intrinsic threshold is given for every node. Which of the following statements is INCORRECT for this case? **1 point**

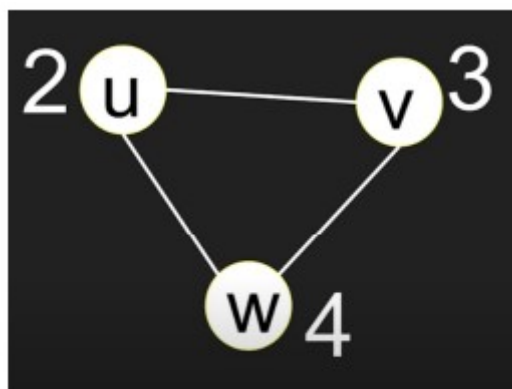


Figure 16: Collective Action

- ☐ u will go for a protest as it has two neighbors and its threshold is two



unit=121&lesson=136)

● Quiz: Week 7 : Assignment (assessment? name=254)

Week 8 ()

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- ☐ v will not go for a protest because w has a threshold of 4 and there are only two other nodes in the network
- ☐ w will not go for a protest because its threshold is 4 and there are only two neighbors
- ☒ all three nodes will go for a protest as the average threshold is 3 and there are three nodes in the network

8) Given a node X having 10 friends/neighbors. 3 of its neighbors have decided to adopt the behavior/action A having a payoff of 12 while 7 of its friends have adopted the action B yielding a payoff of 9. What is (i) the payoff that X gets from its friends who have adopted the action A, (ii) the payoff that X gets from its friends who have adopted B, (iii) The final action/behavior adopted by X? **1 point**

- ☒ (i) 36 ,(ii) 63, (iii) B
- ☐ (i) 63 ,(ii) 36, (iii) B
- ☐ (i) 36 ,(ii) 63, (iii) A
- ☐ (i) 63 ,(ii) 36, (iii) A

9) Consider the social network in the Figure 17, and again let's suppose that in the A-B coordination game. Initially, all nodes have adopted B. There are two initial adopters node 7 and 8 for A. we have a payoff of $a = 3$ and $b = 2$, what is the threshold for any node to switch from B to A? **1 point**

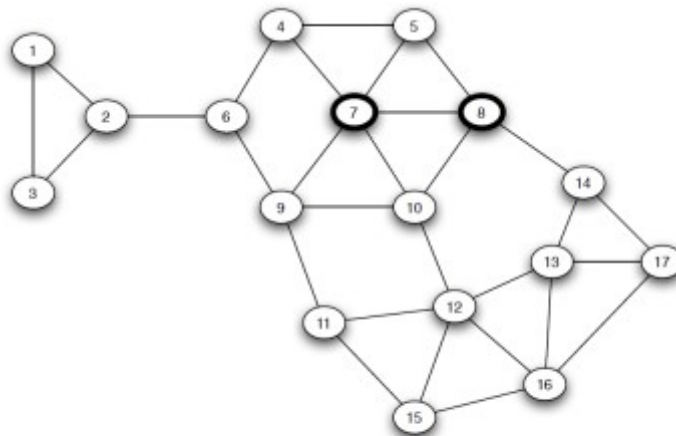


Figure 17: A-B Coordination Game

- ☐ 2/3
- ☐ 3/2
- ☒ 2/5
- ☐ 3/5

10) For the A-B coordination Game mentioned in Question 9, What are the nodes that have adopted action A after the next three steps? **1 point**

- ☒ nodes 5 to 10
- ☐ nodes 4 to 10
- ☐ nodes 4 to 17
- ☐ all nodes 1 to 17



You may submit any number of times before the due date. The final submission will be considered for grading.

Submit Answers

